

Compute Complementarity and Competitive Spillovers: The Stock Market Pricing of Large Language Model Releases

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Abstract

How do stock markets price the release of large language models? We construct a reproducible event-study pipeline covering 124 day-verified model releases from 2022 to 2026 and an exogenously selected universe of Nasdaq-100 and PHLX Semiconductor index constituents, with ecosystem positions assigned by double-blind coding. Three results emerge. First, upstream hardware suppliers earn 1.7 percentage points and direct competitors 1.6 percentage points of cumulative abnormal return over twenty trading days after a release; downstream deployers show a precisely estimated zero. Second, the hardware premium is a late-sample phenomenon: absent through 2024, it reaches 4.3 percentage points in 2025–2026 and 4.6 percentage points around reasoning-model releases, consistent with markets learning to treat releases as compute-demand news. Third, model capability itself is not positively priced—anticipated flagship releases earn no capability premium—and open-weight releases do not erode the hardware premium. Earlier results

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built on non-reproducible measurements reverse under our pipeline.

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1. Introduction

On January 27, 2025, one week after the Chinese startup DeepSeek posted the weights of its reasoning model R1 on GitHub, NVIDIA lost 17% of its market value in a single session—the largest one-day dollar loss for any stock in U.S. history. The episode is routinely cited as evidence that artificial intelligence model releases redistribute value along the AI supply chain, with efficient open-source models threatening the rents of compute suppliers. Single events, however, make treacherous evidence. Across the 124 verified model releases we study, the DeepSeek episode sits in the far left tail of the distribution of hardware-supplier returns: on average, upstream hardware firms *gain* after model releases, and they gain most after precisely the kind of reasoning-model releases that R1 exemplifies.

This paper asks how equity markets price large language model (LLM) releases, using the cross-section of ecosystem positions—hardware suppliers, cloud providers, downstream integrators and deployers, competitors, investors, and publishers—to identify which economic channel the market capitalizes. We assemble 124 release events between April 2022 and March 2026 whose announcement dates we verify to the day against first-party sources, and we measure abnormal returns for 108 index-member firms whose ecosystem positions are assigned under a double-blind coding protocol (Cohen’s κ

between 0.84 and 1.00 across dimensions). Because the firm universe consists of Nasdaq-100 and PHLX Semiconductor constituents, selection into the sample is exogenous to any hypothesis about AI exposure, and the thirty-one constituents with no classifiable relationship to AI provide a within-event control group.

Three findings emerge. First, releases redistribute value toward the upstream and toward rivals. Over a $[0, +20]$ trading-day window, hardware suppliers earn a cumulative abnormal return (CAR) of 1.66 pp ($p = 0.012$; 1.79 pp under a Fama–French three-factor benchmark, $p = 0.007$) and direct competitors earn 1.61 pp ($p = 0.013$), relative to unrelated firms in the same event window and conditional on firm characteristics. Downstream deployers—firms that use AI as a tool inside a non-AI business—show a CAR of -0.07 pp with a standard error of 0.25 pp: a precisely estimated zero that rules out effects more negative than -0.6 pp. The same contrast holds in trading activity. Hardware suppliers display abnormal volume of 0.12 standard deviations on the announcement day itself ($p < 0.01$), while deployers show neither abnormal returns nor abnormal volume; cloud providers trade heavily (abnormal volume 0.29 standard deviations, $p < 0.01$) without directional price consensus.

Second, the hardware premium is not a constant feature of the AI era; it *formed* during the sample. Around LLM releases in 2022–2024 the within-event hardware coefficient is statistically zero. In 2025–2026 it is 4.28 pp ($p < 0.001$), and around reasoning-model releases it reaches 4.61 pp. The window profile in the late sample shows textbook gradual incorporation: significant by day two, accumulating monotonically through day twenty, orthogonal to

the pre-event drift. Even the scope of the pricing rule expanded: releases of media-generation models moved hardware stocks *negatively* in 2024 (-4.7 pp, $p = 0.03$) but positively in 2025–2026 ($+3.5$ pp, $p = 0.02$), as video generation itself became compute-intensive. We interpret this as the market updating its model of what an LLM release means: from application news to compute-demand news.

Third, what the market prices is position, not capability. Using Artificial Analysis benchmark scores as a direct measure of model capability, we find no positive capability premium: among closed-source releases the capability slope is significantly *negative* (-0.06 pp per index point, $p = 0.02$), consistent with flagship releases being priced in advance of the announcement; among unrelated firms the slope is zero, as a placebo requires. Capability matters only as an amplifier of position: the interaction of capability with the hardware indicator is positive. Relatedly, open-weight releases do not attenuate the hardware premium (interaction $p = 0.90$)—the market, unlike the commentary that followed the DeepSeek episode, does not price open weights as a threat to compute rents on average.

The magnitudes are economically material. A 4.3 pp within-event spread over twenty trading days, applied to hardware constituents whose market capitalizations range from tens of billions to several trillion dollars, represents tens of billions of dollars of relative revaluation per flagship release. At the portfolio level, a calendar-time strategy long the thirty hardware suppliers and short the forty downstream firms earns a three-factor alpha of about 24% annualized over the sample ($p = 0.054$); because event windows blanket 56% of trading days, we read the portfolio as scale rather than identification

and place the identifying weight on within-event comparisons.

Our identification strategy rests on within-event cross-sectional comparisons. Event fixed effects absorb everything common to an event window, including market-wide AI sentiment; position indicators are pre-assigned at the (firm, developer) level under a codebook fixed before estimation; and the thirty-one zero-relationship constituents anchor the counterfactual. The design survives an extensive battery: the hardware and competitor premia are significant in 16–17 of 20 specifications spanning benchmarks (S&P 500, Nasdaq-100, SOX, three-factor), windows, winsorization, date-confidence subsamples, exclusion of multi-component events, and event fixed effects, and they are insensitive to release-time corrections (we verify announcement timestamps for 55 events; shifting all events by one trading day leaves the $[0, +20]$ estimate unchanged at 4.68 pp in the late sample).

A final contribution is methodological. An earlier version of this project, built—as much of the emerging literature is—on hand-collected events and externally supplied abnormal returns, found the opposite headline results: a significant *negative* effect for downstream firms and a significant *positive* capability premium for closed-source releases. Neither survives the reproducible pipeline. A forensic decomposition shows why: 17 of 24 matched legacy events carried announcement dates 2 to 120 days off first-party evidence; the legacy “ $[0, +20]$ ” returns behave as if accumulated over roughly $[-10, +20]$; and under *any* date–window combination the legacy returns correlate with assumption-free market-adjusted returns at no more than 0.55, against 0.93 for our measure. Since results this fragile can steer an entire narrative, we report the full specification curve and release the pipeline.

We contribute to three literatures. Event studies of technology announcements document value effects for innovators and rivals (Chaney et al., 1991; Chen et al., 2005; Kogan et al., 2017); we extend the object of study from the announcing firm to the full ecosystem cross-section, and we show the pricing rule itself changed over a four-year window. Work on AI and asset prices ranks firms by exposure to generative AI (Eisfeldt et al., 2023; Babina et al., 2024); our position taxonomy unbundles “exposure” into channels with opposite predictions and shows that the compute-complementarity channel, not average exposure, carries the pricing. Studies of information transfer along supply chains (Lang and Stulz, 1992; Cohen and Frazzini, 2008; Gofman et al., 2020; Barrot and Sauvagnat, 2016) predict heterogeneous spillovers from economic linkages; the positive competitor spillover we find indicates that contagion-type industry information dominates the competitive effect (Lang and Stulz, 1992) in a fast-growing technology sector. Methodologically, we join the literature on specification fragility and researcher degrees of freedom (Simonsohn et al., 2020; Brodeur et al., 2020; Mitton, 2022) with a concrete case in which measurement reconstruction reverses published-style results.

Section 2 develops hypotheses. Section 3 describes the pipeline, the coding protocol, and the measurement audit. Section 4 presents baseline results, Section 5 the formation of compute pricing, Section 6 the capability tests, and Section 7 robustness. Section 8 concludes.

2. Related literature and hypotheses

2.1. The economics of an LLM release

A frontier model release is informative about cash flows at several layers of a vertical supply chain at once. At the top sit the model developers themselves—some listed (Alphabet, Meta, Microsoft, Amazon), most private (OpenAI, Anthropic, xAI, DeepSeek). One layer down are the compute complements: designers of accelerators and high-bandwidth memory, the foundries and equipment makers behind them, and the interconnect, power, and server vendors that turn chips into clusters. Cloud platforms rent this capacity and increasingly resell frontier models through hosted interfaces. Downstream, integrators build products whose core input is model capability, while deployers use AI as a tool inside businesses whose revenue does not depend on it.

Which layer a release is news for depends on how models turn into costs and revenues. Training is a one-off, capitalized expenditure; inference is a recurring marginal cost that scales with usage. Two developments during our sample window shifted the expected compute intensity of the industry from the first margin to the second. Reasoning models—beginning with releases in late 2024 and accelerating through 2025—consume orders of magnitude more inference compute per query by trading test-time computation for accuracy, and their adoption converts model quality improvements into recurring demand for accelerators and data-center capacity. Video-generation models underwent the same transition in 2025, when high-resolution synthesis became a mass-market inference workload. If investors understand this mapping, a capability release is derived-demand news for the upstream; if they instead

read releases as application or displacement news, the pricing should concentrate downstream. The cross-section of positions around releases therefore identifies which economic model of AI the market holds at a point in time—and letting the coefficients vary over time reveals whether that model itself changed.

2.2. Predictions

Classical event-study logic holds that publicly datable, salient events allow abnormal returns to measure revisions in expected cash flows (Fama et al., 1969; Brown and Warner, 1985; MacKinlay, 1997; Kothari and Warner, 2007). Product introductions carry such information for the announcer and, with opposite sign, for rivals (Chaney et al., 1991; Hendricks and Singhal, 1997; Chen et al., 2005; Sorescu et al., 2007). An LLM release differs from a conventional product introduction in that it simultaneously shifts the expected cash flows of firms across an entire vertical ecosystem, as just described.

Two families of predictions follow. The *complementarity* view treats a frontier release as news about derived demand for training and inference inputs: more capable models require and monetize more compute, benefiting upstream asset owners (Acemoglu et al., 2012; Gofman et al., 2020). The *substitution* view emphasizes that as foundation models absorb capabilities, the differentiation space of downstream products narrows (Bommasani et al., 2021; Zhu and Liu, 2018), depressing deployer and integrator values. Industry information transfer adds a third margin: a release may signal sector-wide growth (contagion) or a rival’s competitive loss (Lang and Stulz, 1992). Finally, appropriability logic suggests open-weight releases should

weaken any premium built on proprietary compute demand (Teece, 1986; Lerner and Tirole, 2002; Solaiman, 2023).

We test four hypotheses.

H1 (*position heterogeneity*): abnormal returns around LLM releases differ systematically across ecosystem positions within the same event.

H2 (*direction*): upstream compute complements earn positive abnormal returns; downstream deployers earn negative abnormal returns. (We find the first half supported and the second half rejected in favor of a precise zero.)

H3 (*open-source moderation*): the upstream premium is weaker for open-weight releases. (Not supported.)

H4 (*formation*): if the premium reflects compute-demand news, its strength should increase as the reasoning paradigm shifts compute from one-off training toward recurring inference; the pricing rule should strengthen over time and be strongest around reasoning-model releases. (Supported, including an expansion of the rule’s scope to compute-intensive media models.)

3. Data and measurement

3.1. Release events

The event universe intersects two independent public sources: the AI Timeline project, a curated chronology of AI model releases, and Artificial Analysis (AA), a third-party benchmarking platform. An event enters the candidate set only if the released model is covered by AA, which screens out announcements too minor to be benchmarked. Identity matching between the two sources is resolved by a scored matcher with all manual decisions recorded in an auditable decision table; family releases announced together

are merged into one event. The pipeline yields 134 events, of which 125 have day-level release dates verified against first-party sources (vendor engineering blogs, release notes, model repositories, or archived first captures); the nine events without verifiable day-level dates are dropped. One further event whose identity is inconsistent across sources is flagged and excluded from the main sample, leaving 124 events: 3 from 2022, 8 from 2023, 45 from 2024, 59 from 2025, and 9 from 2026. Eighty-seven events are language-model releases with AA intelligence scores; thirty-five are media-generation releases with AA Elo scores; two are speech models without comparable scores.

Date verification matters. For each event we record the source and confidence of the date; known pitfalls include vendor pages that render UTC timestamps (shifting evening announcements to the next calendar day) and Chinese vendors’ Beijing-date pages (which typically coincide with the correct U.S. reaction day). For 55 events we further recover intraday release times from page metadata or archived first captures; only two events have positive evidence of after-hours release. Section 7 shows results are insensitive to any feasible re-dating.

3.2. Firm universe and ecosystem positions

The firm universe is the union of Nasdaq-100 and PHLX Semiconductor (SOX) constituents at collection date: 108 firms, of which 101 Nasdaq-100 members form the main sample and SOX-only members are used in robustness. Index membership makes selection exogenous to AI exposure and supplies a natural control group: thirty-one constituents (consumer staples, utilities, health care) have no classifiable relationship to any model developer.

Each (firm, developer) pair—108 firms $\times$ 25 developers—is coded on eight

dimensions following a fixed codebook: upstream hardware (R1), upstream cloud (R2), downstream integrator (R3), downstream deployer (R4), downstream enabler (R5), competitor (R6), investor (F1), and owner/publisher (F2). Two coders worked double-blind; agreement was 99.0% across the 21,600 cell decisions, with dimension-level Cohen’s κ between 0.839 and 1.000; disagreements were adjudicated against the codebook with all decisions logged. Event-level indicators take the union over the event’s developers. Event attributes (open-weight status, reasoning, modality, family release) were likewise double-coded (κ 0.917–1.000).

3.3. Abnormal returns and abnormal volume

Daily total returns for all firms and benchmarks come from adjusted closing prices (January 2021 to April 2026). Abnormal returns follow the market model with the S&P 500 as market proxy, estimated over trading days $[-200, -10]$ relative to the event day (minimum 120 observations); we cumulate over $[0, +w]$ for $w \in \{0, 1, 2, 3, 5, 10, 15, 20\}$ and a pre-event window $[-10, -2]$. We deliberately avoid a Nasdaq-100 benchmark for the main results because sample firms are its constituents; QQQ- and SOX-benchmark CARs and Fama–French three-factor CARs are computed for robustness. Firm controls (log assets, book-to-market, prior-year momentum, estimation-window volatility) are matched under a strict as-of rule: only statements whose fiscal period ended at least 45 days before the event date are used. Negative book equity (buyback-driven, 711 firm-events) is flagged and excluded from the book-to-market control rather than entered as a negative ratio.

Abnormal volume is the standardized abnormal log volume: the deviation

of daily log volume from its $[-200, -10]$ mean, scaled by its estimation-window standard deviation, averaged over the event window.

3.4. A measurement audit

An earlier vintage of this project used hand-collected events and externally computed abnormal returns, a practice common in the emerging AI event-study literature. Its two headline results—significantly negative deployer CARs and a significantly positive capability premium for closed-source releases—*reverse* under the pipeline of Sections 3.1–3.3. A three-layer forensic audit locates the difference entirely in measurement rather than in sample composition: restricting the legacy panel to the firms and events shared with the new sample leaves the legacy results intact, while recomputing returns for the identical cells eliminates them.

The three layers are: (i) *dates*—17 of 24 matched legacy events carry announcement dates that differ from first-party evidence by 2 to 120 days; (ii) *windows*—a scan across all window definitions shows the legacy “day $[0, +1]$ ” return series is most consistent with a $[-1, +1]$ window and the legacy “ $[0, +20]$ ” series with roughly $[-10, +20]$, i.e., post-event windows contaminated by pre-event returns; and (iii) an *unattributable residual*—under the best date–window combination the legacy series correlates with assumption-free market-adjusted returns (raw return minus S&P 500 return, no estimation) at 0.55, against 0.93 for our series on the same cells. A two-by-two swap test on identical firm–event cells shows that replacing the capability regressor (old versus new benchmark vintages) leaves the legacy result intact, while replacing the return measure eliminates it. We conclude the legacy results were artifacts of compound measurement error, and we flag this as a caution

for a literature in which event dates are frequently taken from secondary aggregators (Simonsohn et al., 2020; Brodeur et al., 2020; Mitton, 2022).

3.5. Descriptive statistics

Table 1 summarizes the panel: 12,112 firm-event observations (124 events $\times$ 101 firms, less late listings without full estimation windows). Twenty-five percent of firm-events carry a hardware relationship, 5% cloud, 12% integrator, 20% deployer, 5% competitor; 26% of observations belong to zero-relationship control firms.

4. Baseline results: position pricing

4.1. Position premia

Table 2 reports the baseline cross-sectional regressions of $CAR[0, +20]$ on the seven position indicators, firm controls, and year fixed effects, with standard errors clustered by event (CR2). Column (1) uses the S&P 500 market model, column (2) the three-factor model, and column (3) adds event fixed effects, identifying purely from within-event variation.

Hardware suppliers earn 1.66 pp more than unrelated constituents over the twenty days following a release ($p = 0.012$); the estimate is 1.79 pp under the three-factor benchmark and 1.54 pp within events. Direct competitors earn a statistically indistinguishable premium of 1.61 pp. The competitor premium is noteworthy: rivalry-based models of product announcements predict negative wealth effects for competitors (Chen et al., 2005); the positive sign indicates that in this period an LLM release conveyed industry-level growth information that dominated the competitive-displacement effect, the contagion pattern of Lang and Stulz (1992).

Table 1: Sample composition and descriptive statistics

	Events by year and type			
	2022–23	2024	2025	2026
All events	11	45	59	9
LLM (AA intelligence score)	5	32	44	6 ^a
Media generation (AA Elo)	6	12	14	3
Reasoning-model events	0	4	31	6
Open-weight events	4	20	20	3
	Firm-event panel (main sample)			
	Obs.	Mean β	Mean CAR[0, 20]	SD
Upstream hardware	2,708	1.99	+2.08%	12.7%
Upstream cloud	568	1.37	−0.75%	9.6%
Downstream integrator	1,981	1.42	−0.63%	11.3%
Downstream deployer	2,971	0.96	−0.07%	9.3%
Competitor	753	1.24	−0.32%	7.0%
No relationship (control)	3,967	0.68	−0.08%	9.7%

Notes: The main sample excludes one event with inconsistent identity across sources. Position categories overlap (multi-label coding); the control row counts firm-events with no positive indicator on any dimension.

^a Two 2026 speech-model events carry no comparable AA score.

The deployer coefficient is -0.07 pp with a standard error of 0.25 pp. This is not an underpowered non-result: the 95% confidence interval, $[-0.56, +0.42]$ pp, excludes deployer losses larger than one third of the hardware gain. The market did not, on average, price LLM releases as bad news for firms that deploy

Table 2: Ecosystem position and cumulative abnormal returns

Dependent variable:	(1) $CAR^{MM}[0, 20]$	(2) $CAR^{FF3}[0, 20]$	(3) $CAR^{MM}[0, 20]$
Upstream hardware	0.0166** (0.0065)	0.0179*** (0.0065)	0.0154** (0.0066)
Upstream cloud	-0.0011 (0.0078)	0.0020 (0.0074)	-0.0018 (0.0076)
Downstream integrator	-0.0051 (0.0049)	-0.0041 (0.0048)	-0.0053 (0.0050)
Downstream deployer	-0.0007 (0.0025)	0.0002 (0.0024)	-0.0011 (0.0025)
Competitor	0.0161** (0.0064)	0.0159*** (0.0058)	0.0158** (0.0064)
Investor	0.0090 (0.0114)	0.0145 (0.0113)	0.0087 (0.0115)
Owner/publisher	0.0152 (0.0138)	0.0151 (0.0135)	0.0148 (0.0139)
Firm controls	Yes	Yes	Yes
Year fixed effects	Yes	Yes	—
Event fixed effects	—	—	Yes
Observations	12,112	12,112	12,112
Events	124	124	124

Notes: Firm controls are log assets, book-to-market (with a negative-equity indicator), estimation-window volatility, and prior-year momentum. Standard errors clustered by event (CR2) in parentheses. *, **, *** denote 10%, 5%, 1% significance.

AI inside conventional businesses.

The remaining positions sharpen the interpretation. Cloud providers—the most direct resellers of model capability—show a price response of zero (column 1: -0.11 pp, $p = 0.89$), a fact we return to below because it coexists with the largest trading response in the panel. Integrators are mildly negative but imprecise (-0.51 pp, $p = 0.30$). Investors and owner-publishers carry positive point estimates of 0.9 and 1.5 pp but few independent clusters (their positive cells concentrate in a handful of developers), so we do not lean on them. The economically meaningful contrast is the roughly 1.7 pp spread between hardware suppliers and deployers within the same event window—firms facing the same release, the same market conditions, and the same AI sentiment, distinguished only by where they sit in the production chain.

Figure 1 traces the window profile. In the full sample the hardware premium emerges at $[0, +10]$ and grows through $[0, +20]$; in the 2025–2026 subsample it is significant from $[0, +2]$ ($+0.51$ pp), accumulates monotonically to $+4.28$ pp, and is preceded by a modest pre-event run-up. Crucially, controlling for the firm’s own pre-event CAR leaves the post-event estimate unchanged (4.28 pp to 4.35 pp), so the accumulation is not a continuation of pre-existing drift.

4.2. Trading activity

Table 3 repeats the design with standardized abnormal log volume as the outcome. Hardware suppliers trade abnormally heavily on the announcement day itself— 0.12 standard deviations at $[0, +1]$ ($p < 0.01$)—including in the full sample where the same-horizon price response is still indistinguishable from zero. Attention arrives immediately; directional consensus forms

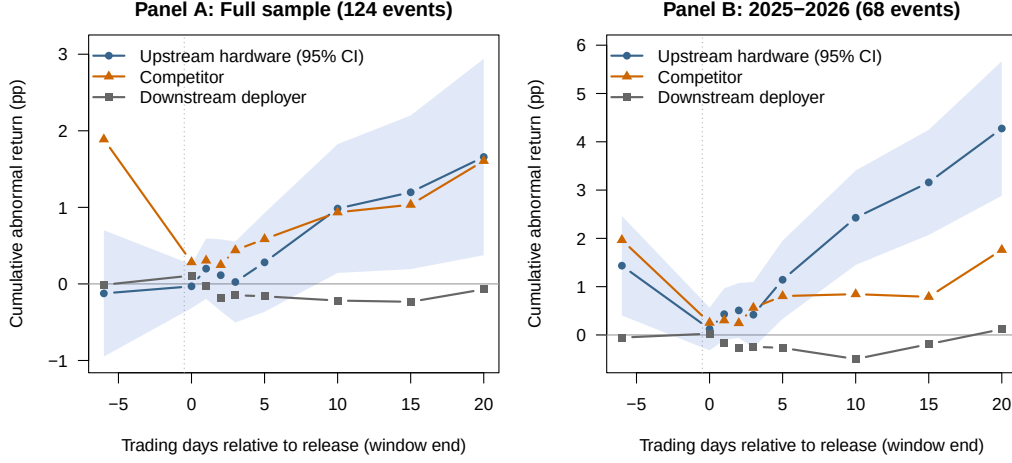


Figure 1: Cumulative abnormal returns around LLM releases by ecosystem position. Coefficients on position indicators from window-by-window cross-sectional regressions with firm controls and year fixed effects; shading is the 95% confidence band for the hardware coefficient (event-level CR2 clustering). The leftmost point is the pre-event window $[-10, -2]$.

over days. Deployers show no abnormal volume at any horizon ($p = 0.74$ at $[0, +1]$): their zero price response reflects genuine irrelevance rather than offsetting disagreement. Cloud providers are the mirror image—the largest abnormal volume in the panel (0.29 standard deviations at $[0, +1]$; 0.44 in 2025–2026) with no directional price response, the signature of repricing activity without consensus. The cloud volume anomaly extends into the pre-event window (0.20 standard deviations over $[-10, -2]$, $p < 10^{-5}$; 0.31 in 2025–2026): cloud names attract heavy trading before and at announcements without ever settling on a direction. In disagreement models, volume without price movement is the fingerprint of investors interpreting the same public signal with different priors (Kandel and Pearson, 1995); cloud plat-

Table 3: Abnormal trading volume around releases

Window:	Full sample		2025–2026	
	$[-10, -2]$	$[0, +1]$	$[-10, -2]$	$[0, +1]$
Upstream hardware	0.033 (0.026)	0.121*** (0.036)	0.046 (0.032)	0.106*** (0.036)
Upstream cloud	0.204*** (0.041)	0.292*** (0.049)	0.314*** (0.046)	0.435*** (0.063)
Downstream deployer	−0.019 (0.015)	0.006 (0.019)	0.024 (0.018)	0.028 (0.021)
Competitor	−0.037 (0.029)	−0.017 (0.043)	−0.014 (0.041)	−0.081 (0.052)
Event fixed effects	Yes	Yes	Yes	Yes

Notes: Outcome is the mean standardized abnormal log volume over the window (deviation from the $[-200, -10]$ mean in estimation-window standard deviations). Firm controls included; standard errors clustered by event.

forms fit naturally, since a frontier release is simultaneously good news for their compute-rental business and bad news for their own model ambitions and gross margins. No other position shows pre-event abnormal volume—in particular, the hardware pre-event price run-up documented above is *not* accompanied by trading intensity. The three downstream fates thus separate cleanly: hardware is priced with consensus, cloud is traded without consensus, and deployers are ignored on both margins.

4.3. *Economic magnitude*

A calendar-time portfolio long the thirty hardware suppliers and short the forty downstream firms earns a three-factor alpha of 9.7 basis points per day over April 2022 to April 2026—roughly 24% annualized ($p = 0.054$, Newey–West). We report this as magnitude rather than as identification: with 124 releases in four years, event windows cover 56% of trading days, so the calendar-time design cannot cleanly separate event-driven from era-driven returns. The within-event estimates of Table 2 carry the identification burden.

5. The formation of compute pricing

5.1. *The premium is a late-sample phenomenon*

Table 4 estimates position coefficients separately by period, with event fixed effects. For LLM releases—the events that carry compute-demand information—the hardware coefficient is a tight zero in 2024 (-1.4 pp, $p = 0.33$) and $+4.3$ pp ($p < 0.001$) in 2025–2026. The formation is monotone for compute-relevant events; there is no intermediate regime.

Two further splits corroborate the interpretation. Around reasoning-model releases—the paradigm that shifts compute from one-off training to recurring inference—the hardware premium is 4.61 pp ($p < 0.001$), with significant responses from day one. And the scope of the rule expanded: media-generation releases moved hardware stocks negatively in 2024 (-4.7 pp) but positively in 2025–2026 ($+3.5$ pp), the period when video generation became a major consumer of inference compute. The market did not merely strengthen

Table 4: The hardware coefficient by period and event type

	LLM releases		Media-generation releases	
	2024	2025–26	2024	2025–26
Upstream hardware	−0.0150 (0.0137)	0.0425*** (0.0088)	−0.0468** (0.0194)	0.0346** (0.0133)
Event fixed effects	Yes	Yes	Yes	Yes
Events	32	50	12	17

Notes: Within-event regressions of $CAR[0, 20]$ on all position indicators and firm controls, estimated separately by period and event type. In 2022–23 (11 events) the LLM cell is +0.3pp ($p = 0.91$, 5 events) and the media cell +4.9pp ($p = 0.02$, 6 events); both rest on too few events to interpret. Standard errors clustered by event.

an existing pricing rule; it extended the rule to a class of events it had previously treated as irrelevant or negative for compute suppliers. Both facts are difficult to reconcile with a constant risk-based explanation and natural under gradual learning about the economic content of releases.

The timing of formation aligns with the arrival of the paradigm it prices. The first reasoning models entered the sample in September 2024; open-weight reasoning followed in January 2025; by mid-2025 reasoning tiers were standard across developers, and 31 of the 59 releases of 2025 in our sample are reasoning models, against 4 of 45 in 2024. The pricing rule strengthened exactly as the inference-intensive share of the release flow rose. We remain deliberately agnostic on whether investors learned from the releases themselves or from the contemporaneous cash-flow evidence of data-center

capital expenditure; distinguishing the two requires expectations data, which we leave to the extension discussed in the conclusion. What the cross-section establishes is narrower but robust: the mapping from release events to upstream revaluation was not in prices before 2025 and is in prices afterward.

5.2. *Anticipation*

In the late sample, hardware suppliers also show a pre-event run-up of 1.4pp over $[-10, -2]$ ($p < 0.01$). Three facts indicate anticipation rather than confounded drift: the run-up is absent in 2022–2024; the post-event estimate is unchanged by controlling for the firm’s pre-event CAR; and there is no abnormal pre-event volume. Frontier releases are rumored, scheduled, and teased; partial pre-announcement pricing is expected, and it implies our post-announcement estimates are lower bounds on the total revaluation.

6. Capability is not priced—position is

If markets priced model capability directly, more capable releases should produce larger revaluations. Table 5 tests this with AA intelligence scores (87 LLM events). The pooled capability slope is zero. Among closed-source releases—where capability rents would be most appropriable (Teece, 1986)—the slope is *negative* (-0.060 pp per index point, $p = 0.024$; -1.3 pp per standard deviation under the three-factor benchmark). Among firms with no ecosystem relationship the slope is an exact zero, as a placebo requires. And the interaction of capability with the hardware indicator is positive ($p = 0.088$): capability amplifies the position premium even though it carries no average price.

Table 5: Model capability and abnormal returns

	(1) All	(2) Closed	(3) Open	(4) Placebo (unrelated firms)	(5) Interaction
AA intelligence	−0.0001 (0.0002)	−0.0006** (0.0002)	0.0000 (0.0003)	−0.0001 (0.0004)	−0.0003 (0.0002)
× Upstream hardware					0.0010* (0.0006)
Upstream hardware					0.0207*** (0.0078)
Events	87	50	37	87	87

Notes: Dependent variable $CAR^{MM}[0, 20]$. All columns include firm controls and year fixed effects; standard errors clustered by event (CR2). Capability is centered in column (5).

The negative closed-source slope has a natural reading: within a period, the most capable releases are flagship events—rumored for months, dated in advance—whose information reaches prices before the announcement window, while less anticipated releases deliver larger surprises. Consistent with this, the highest-capability closed releases in our sample all carry positive average CARs; the negative slope is a within-year, relative-anticipation pattern. Replacing the capability level with a surprise measure—the score’s distance above the pre-event frontier maximum—does not rescue a positive slope (Appendix Appendix E), because benchmark-measured surprise still correlates with exactly the flagship status that markets anticipate; separating true expectation errors from levels requires ex-ante forecast data that do not yet exist for model benchmarks. Two conclusions follow. First, “capa-

bility exposure” is not a priced characteristic around releases; what is priced is the release’s implication for ecosystem positions. Second, the open-weight distinction that dominates public commentary carries no average price: the hardware $\times$ open-weight interaction is 0.16 pp with $p = 0.90$ (Section 7), and open releases show the same zero capability slope as closed ones.

7. Robustness

Table 6 condenses the specification battery. The hardware and competitor premia are significant in 16 and 17 of 20 specifications, respectively; the deployer zero and the null open-weight interaction hold in all 20. Highlights: (i) *benchmarks*—QQQ, SOX, and three-factor CARs give hardware estimates of 1.6, 1.1, and 1.8 pp; the SOX-benchmark estimate is within-sector and still positive; (ii) *windows*— $[0, 10]$ and $[0, 15]$ give 1.0 and 1.2 pp; (iii) *dates*—restricting to high-confidence dates leaves the point estimate at 1.2 pp (significance drops with 104 events); excluding multi-component events, 1.5 pp; (iv) *event-day definition*—applying evidence-based after-hours corrections (two events) or shifting *all* events one trading day leaves the late-sample estimate at 4.1–4.7 pp; (v) *winsorization* at 1%/99%, 1.7 pp; (vi) *event fixed effects*, 1.5 pp; (vii) *pre-event CAR control*, unchanged at 1.7 pp; (viii) *composition*—the premium is not a single-stock artifact: dropping any one hardware firm leaves the coefficient between 1.2 and 1.6 pp, excluding NVIDIA gives 1.5 pp ($p = 0.023$), and non-SOX hardware names (servers, networking, storage) earn at least as much as SOX members (Appendix Appendix E). A specification curve over 3,144 estimates spanning outcomes, samples, moderators, and fixed-effect structures shows the hardware–time interaction positive in 100%

Table 6: Robustness summary (hardware coefficient, CAR[0, 20])

Specification	Coef.	p	Specification	Coef.	p
Baseline (S&P 500 MM)	0.017	0.01	Excl. multi-component	0.015	0.02
FF3	0.018	0.01	High-confidence dates	0.012	0.13
QQQ benchmark	0.016	0.01	Event fixed effects	0.015	0.02
SOX benchmark	0.011	0.01	Winsorized 1/99	0.017	0.01
Window [0, 10]	0.010	0.02	Pre-event CAR control	0.017	0.01
Window [0, 15]	0.012	0.02	All events shifted $t+1^a$	0.047	0.00
LLM events only	0.020	0.01	Reasoning events only	0.046	0.00
2025–26 only	0.043	0.00	2022–24 only	−0.012	0.47
Media events only	0.010	0.44	Chinese-developer events	0.027	0.06

Notes: Each cell is the upstream-hardware coefficient from the baseline specification with the stated modification.

^a 2025–26 subsample.

of specifications (family-wise $q = 0.002$); the corresponding downstream interactions do not survive joint conditioning on all position dimensions with phase-specific coefficients.

8. Conclusion

Between 2022 and 2026, U.S. equity markets developed a rule for pricing large language model releases. The rule they converged on is not the one implied by either popular narrative. Releases are not priced as broad AI enthusiasm: only firms positioned as compute complements or direct rivals revalue, while the average constituent—and, notably, the average firm deploying AI in a conventional business—shows neither price nor volume response. Nor

are releases priced as capability milestones: benchmark scores carry no positive premium, and open-weight releases do not dent the hardware premium. What the market prices is structural position, and it learned to do so during the sample—the hardware premium was absent through 2024, formed with the reasoning paradigm in 2025, and extended to compute-intensive media models thereafter.

For investors, the results imply that release calendars are information events for a narrow, identifiable cross-section, and that the tradable content of a release lies in supply-chain position rather than in headline capability. For the debate on AI and firm values, the precise deployer zero is a discipline on displacement narratives: whatever the long-run effects of foundation models on downstream businesses, four years of release events contain no evidence that markets price displacement on announcement. For empirical work on AI events, our measurement audit is a caution: two headline results of an earlier vintage of this project reversed once dates and returns were rebuilt reproducibly, and the reversal was attributable to compound measurement error invisible without a full audit trail.

The natural next margin is expectations data: analyst reports and earnings calls can date when compute-demand language entered the sell-side lexicon, testing the learning interpretation directly. A second is attention: media-coverage weights would separate salient releases from routine ones and sharpen the anticipation mechanism we document among flagship models.

Data availability

The event pipeline, coding decision tables, and estimation code are available from the authors; all release dates and position codes carry documented first-party evidence.

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Online Appendix

Appendix A. Event sample construction and dating

Appendix A.1. Pipeline

The candidate universe is AiTimeline (ATL), a community-maintained release chronology organized by year-month. Candidates are matched to Artificial Analysis (AA) records scored on name similarity, developer, modality, and version; all manual match adjudications (136 entity-level decisions with reasons) are recorded in an entity-level decision file in the replication package. Known failure modes of string matching documented during the audit: version-boundary false positives (“Claude 3” $\subset$ “Claude 3.5” scores perfectly) and tie-breaking by list order; both caused misses of flagship events (GPT-5, Claude Sonnet 4.5) in early passes and were fixed by decision-file overrides. Sample rules fixed by the authors: (i) family releases announced together form one event; (ii) an event qualifies if AA covers *any* variant tier; (iii) when an announcement and a release both appear, the release is kept. The intersection yields 134 events covering April 2022–March 2026.

Appendix A.2. Day-level dating

Every event date comes from a first-party source with URL and confidence grade in the dating decision file of the replication package: vendor engineering blogs and release notes (preferred), official documentation or model repositories, or first-party technical reports. Where pages expose no date, the earliest Wayback Machine capture of the official page dates it (e.g., DALL-E 2 at 2022-04-06, Stable Diffusion 1.5 at 2022-10-20, the GPT-4o-0806 structured outputs post at 2024-08-06). Multi-component events take the

earliest component date, with all component dates preserved. Two systematic timezone hazards are documented: vendor CMS timestamps rendered in UTC can postdate a U.S. afternoon announcement by one calendar day (the Claude 3.5 Sonnet page reads 2024-06-21T03:28Z; the announcement was June 20 U.S.); and Chinese vendors publish under Beijing dates, which typically coincide with the first U.S. trading day able to react, requiring no adjustment. Dating revisions made after first-pass verification: Claude 3.5 Sonnet 06-21→06-20 (UTC artifact); Imagen 3 re-dated to the technical report (2024-08-13) after the first pass had cited an unrelated December blog; Gemini 1.5 Flash-8B re-dated to the experimental-release changelog entry (2024-08-27); Claude 3.5 Haiku re-dated to availability (2024-11-04, platform release notes) rather than the October pre-announcement.

Appendix A.3. Excluded events

Nine of 134 events lack any day-level first-party date and are dropped: Kling 1.5, Kling 1.6, Pika 1.5, Pika 2.0, Wan 2.5 Preview, the Z-Image-Turbo/Qwen-Image bundle, Gemini 3 Deep Think (the dated first-party post covers the initial December 2025 launch, not the February 2026 update ATL records), and two month-only events (Firefly Image 3, Seedance 2.0). One further event (Grok 4.20) is excluded from the main sample because ATL month (2026-02), the vendor release note (2026-03-10), and the AA snapshot (2026-04-07) disagree about its identity; it carries an exclusion flag in the panel. Main sample: 124 events.

Appendix A.4. Release timing

Publication timestamps were collected for 55 events: 41 from live page metadata (23 during U.S. market hours, 9 pre-market, 2 after-hours, 7 dated the prior day—these last mostly Beijing-midnight date-only stamps) and 14 from Wayback first-capture upper bounds. Two events are documented after-hours: Grok 4 (late-evening PT livestream, 2025-07-09) and Phi-4 Multimodal (17:00 ET). Metadata caveats recorded: Anthropic CMS timestamps can lag the actual announcement (the 3.5 Sonnet stamp reads 23:28 ET against midday press coverage), and Chinese vendors’ stamps are date-only. Table A.7 shows the 2025–26 hardware coefficient under three day-boundary conventions.

Table A.7: Event-day conventions: hardware coefficient, 2025–2026

Window	Original dates	Evidence shifts (2 events)	All events +1 day
[0, 1]	0.0044 (0.114)	0.0041 (0.146)	0.0037 (0.126)
[0, 2]	0.0050 (0.090)	0.0046 (0.119)	0.0025 (0.403)
[0, 5]	0.0112 (0.007)	0.0109 (0.009)	0.0111 (0.004)
[0, 20]	0.0409 (0.000)	0.0406 (0.000)	0.0468 (0.000)

p-values in parentheses; event fixed effects, CR2.

Appendix B. Firm universe and double-blind coding

Appendix B.1. Universe

108 firms: 101 Nasdaq-100 constituents (main) plus 7 SOX-only names (robustness), pulled at collection with source provenance in the replication

package. Known limitation: the list is a point-in-time snapshot, not a historical reconstitution, so survivorship enters only through index membership; a robustness excluding post-2024 index additions is available in the pipeline. No IT-services firm (Accenture, IBM, Infosys) belongs to either index, so the downstream-enabler dimension (R5) is structurally empty in this universe—a difference from the legacy hand-picked sample that any comparison must respect.

Appendix B.2. Relationship coding

Each (firm, developer) pair ($108 \times 25 = 2,700$) is coded on R1 hardware, R2 cloud, R3 integrator, R4 deployer, R5 enabler, R6 competitor, F1 investor, F2 owner, per a written codebook (v1.0) with decision procedure and precedent tables. Two coders worked independently; cell-level agreement was 99.0%. Dimension-level κ : R1 1.00, R2 1.00, R3 0.96, R4 0.84, R5 1.00, R6 0.85, F1 1.00, F2 1.00. The 213 disagreement cells reduced to 12 firm-level questions, adjudicated against the legacy sample’s adjudication records where available (Microsoft carries R3 via Copilot; large platform developers are competitors to all developers; Cisco is R4-only; Meta carries no R4; SBUX deploys via a named program; IDEXX’s AI diagnostics qualify; MAR/PCAR fail the evidence bar; Adobe is not a competitor to LLM-first labs). Five codebook extensions surfaced by the new universe are flagged for future recoding rounds: semiconductor-equipment makers coded R1 at confidence M (ASML, AMAT, LRCX, KLAC, TER, ENTG); power utilities carry no dimension (CEG’s data-center nuclear PPA is the clearest omission); Cisco’s classification follows legacy precedent over the codebook letter; NVIDIA’s 2025 investments in OpenAI and xAI make F1 time-varying

(coded at L); Alibaba has no listed owner in this universe, so its eight events carry no F2 row. Codebook rule preserved: NVIDIA is never coded R6—its exposure is the hardware channel.

Appendix B.3. Event labels

Nine event-level fields (modalities, cross-modality, family, multimodal, reasoning, coding-focused, media-generation, open-weight, Chinese-developer) were double-coded under the same protocol: agreement 99.0%, κ by field 0.92–1.00. Ten disagreements were adjudicated, six with hard evidence (e.g., Kimi K2.5’s open weights and vision confirmed from its repository card). One rule amendment is logged: open-weight is judged at the event level (any principal variant with public weights), because flagship-tier judging misclassifies families like FLUX whose API flagship coexists with open-weight variants. Judgment convention: hybrid-tier models count as reasoning when the AA representative record is a reasoning tier; capability representatives are the highest Intelligence Index (LLM) or arena score (media) component—the “flagship rule.”

Appendix C. Panel construction and design decisions

Appendix C.1. Outcome verification

CARs are computed by scripted market models; an independent re-implementation reproduces α , β , and CARs to six decimals on audit cells, and the R1-event NVIDIA cell captures the documented 2025-01-27 crash (-5.2% over $[0, +5]$). Price data checks: zero duplicate (date, symbol) rows; zero missing trading days against the SPY calendar; splits verified (NVDA 10:1); all nine

$|\text{return}| > 40\%$ observations correspond to real earnings or trial events. Betas are economically ordered (utilities ≈ 0 , NVDA ≈ 2).

Appendix C.2. Fundamentals

Statement data merge under a lagged as-of rule: the most recent fiscal period whose end predates the event by at least 45 days (filing lag), capped at 400 days of staleness (accommodating semiannual reporters CCEP and FER). The first build used same-quarter matching, under which 86% of rows carried balance sheets not yet closed at the event date; the as-of rule eliminates all look-ahead (median statement age 87 days). Negative book equity (buyback-driven: BKNG, SBUX, MAR, ORLY, MSTR, among others; 711 firm-events) sets book-to-market to missing with a dummy; the raw value is retained. ADR share counts are verified against ADS ratios (TSM 5:1, ASX 2:1, UMC 5:1). AA capability metrics are collection-time snapshots, not release-vintage values—consistent with the legacy convention, absorbed by year effects, and immaterial to the capability results (Appendix D swaps vintages).

Appendix C.3. Design decisions log

(P4) SPY is the market proxy because the sample is the Nasdaq-100 (QQQ self-benchmarks; retained as robustness). (D1) Grok 4.20 excluded for identity conflict. (D2) Multi-component events keep the earliest component date; excluding the six such events is a robustness row. (D3) Day-boundary conventions immaterial (Table A.7). (D4) Late-listed firms require 120 estimation observations; 327 firm-events drop. (D5) Semiannual reporters handled by the staleness cap. (D6) Event-clustered SEs (CR2) throughout; event

fixed effects as robustness. (D7) Estimation window $[-200, -10]$.

Appendix D. Measurement forensics: why the legacy results reverse

The legacy project (60 hand-collected events, 86 hand-picked firms, externally supplied CARs) reported a deployer loss of -1.9pp and a positive closed-source capability slope ($+0.23\text{pp}$ per index point, wild-bootstrap $p = 0.008$). Neither replicates. The decomposition proceeds in three layers.

Layer 1: not the sample. Within the legacy panel (its own CARs, dates, and capability vintages), the closed-source capability slope survives every composition cut: all firms $+0.232$ ($p = 0.007$); only firms also in the new universe $+0.193$ ($p = 0.0003$); only firms *not* in the new universe $+0.252$ ($p = 0.018$); only events matched to the new sample $+0.211$ ($p = 0.069$); the minimal intersection $+0.192$ ($p = 0.007$).

Layer 2: the measurement layer. On the same intersections computed with reproducible CARs and verified dates, every cell is zero or negative: full new panel -0.060 ($p = 0.024$); overlap firms -0.075 ($p = 0.084$); matched events -0.096 ($p = 0.097$); minimal intersection -0.019 ($p = 0.82$). The significantly negative slope is contributed by 2025–26 events absent from the legacy sample; the legacy-positive is contributed by the measurement layer.

Layer 3: a 2×2 swap test on 493 identical (event, firm) cells crosses outcome vintage with capability vintage: legacy CAR $\times$ legacy capability $+0.177$ ($p = 0.06$); legacy CAR $\times$ new capability $+0.169$ ($p = 0.12$); new CAR $\times$ legacy capability $+0.010$ ($p = 0.80$); new CAR $\times$ new capability -0.008 ($p = 0.84$). Swapping the capability vintage changes nothing; swap-

ping the outcome kills the result. The legacy positive lives entirely in the legacy abnormal returns.

Diagnosing the legacy outcome. Parsed date comparison on 24 matched events: 7 agree within a day; 17 differ by 2–120 days (Gemini 1.5 entries by 1–2 months; Veo 2 by four months). Against an assumption-free benchmark (raw return minus S&P 500 return, cumulated), on same-date U.S. cells the reproducible CAR correlates 0.93 and the legacy CAR 0.65. A window-definition scan shows the legacy “CAR[0, +1]” best matches $[-1, +1]$ (0.88 vs. 0.63 at its nominal window) and the legacy “CAR[0, +20]” best matches $[-10, +20]$ (0.84 vs. 0.72)—the legacy post-event windows embedded two weeks of pre-event drift, consistent with their larger dispersion (SD 12.8% vs. 10.4%). Even granting the legacy series its own dates and any window, its correlation with observable price movements peaks at 0.55 on the full sample. Two counterfactual checks fail to rescue it: applying the contaminated $[-10, +20]$ window to correct data still yields a negative capability slope (-0.073 , $p = 0.04$), and capability does not predict pre-event run-ups in correct data ($p = 0.58$). The correct characterization is compound measurement error in an unauditable pipeline—misdated events, mislabeled windows, and an unattributable residual—rather than any single correctable mistake.

Appendix E. Additional results

Appendix E.1. Full robustness matrix

Table E.8 reports all twenty specifications.

Table E.8: Robustness matrix, all rows (hardware / deployer / competitor coefficients)

Specification	Events	Hardware	Deployer	Competitor
S01 Baseline SPY [0, 20]	124	0.0166**	−0.0007	0.0161**
S02 + open-weight interaction	124	0.0160**	−0.0007	0.0160**
S03 QQQ benchmark	124	0.0162**	0.0000	0.0147**
S04 SOXX benchmark	124	0.0112**	−0.0003	0.0159**
S05 Fama–French 3F	124	0.0179***	0.0002	0.0159***
S06 Window [0, 10]	124	0.0098**	−0.0022	0.0093**
S07 Window [0, 15]	124	0.0120**	−0.0023	0.0103*
S08 Pre-event CAR control	124	0.0166**	−0.0007	0.0157**
S09 Excl. multi-component	118	0.0154**	−0.0013	0.0165**
S10 High-confidence dates	104	0.0118	−0.0012	0.0153**
S11 2024-04 onward	110	0.0159**	0.0021	0.0210***
S12 2022–2024 only	56	−0.0123	−0.0038	−0.0014
S13 2025–2026 only	68	0.0428***	0.0012	0.0176*
S14 LLM events only	87	0.0197**	−0.0005	0.0196**
S15 Media events only	35	0.0103	−0.0002	0.0084
S16 Event fixed effects	124	0.0154**	−0.0011	0.0158**
S17 Reasoning events only	41	0.0461***	0.0040	0.0241*
S18 Chinese-developer events	25	0.0265*	−0.0053	0.0201
S19 Non-Chinese events	99	0.0148**	0.0005	0.0140**
S20 CAR winsorized 1/99	124	0.0168***	−0.0001	0.0136**
S02 interaction term: hardware $\times$ open = 0.0016 ($p = 0.90$).				

Appendix E.2. Window profiles

Table E.9 gives the coefficients behind Figure 1 plus the reasoning-event panel.

Table E.9: CAR profiles by accumulation window (hardware / competitor / deployer)

Window	Full sample (124)			2025–26 (68)			Reasoning (41)		
	HW	Comp	Depl	HW	Comp	Depl	HW	Comp	Depl
$[-10, -2]$	−0.001	0.019***	−0.000	0.014***	0.020***	−0.000	0.017**	0.027***	−0.003
$[0, 0]$	−0.000	0.003*	0.001**	0.001	0.003	0.000	0.003	0.005**	0.001
$[0, 1]$	0.002	0.003	−0.000	0.004	0.003	−0.002	0.006*	0.005	−0.001
$[0, 2]$	0.001	0.002	−0.002*	0.005*	0.003	−0.003**	0.009**	0.004	−0.003*
$[0, 3]$	0.000	0.004	−0.002	0.004	0.006	−0.002*	0.007	0.009	−0.002
$[0, 5]$	0.003	0.006	−0.002	0.011***	0.008	−0.003	0.015**	0.011	−0.000
$[0, 10]$	0.010**	0.009*	−0.002	0.024***	0.008	−0.005*	0.026***	0.016*	−0.000
$[0, 15]$	0.012**	0.010*	−0.002	0.032***	0.008	−0.002	0.037***	0.016	0.001
$[0, 20]$	0.017**	0.016**	−0.001	0.043***	0.018*	0.001	0.046***	0.024*	0.004

Year fixed effects, CR2 by event. The pre-event run-up coexists with unchanged post-event coefficients under pre-CAR control (2025–26 hardware: 0.0428 $\rightarrow$ 0.0435).

Appendix E.3. Abnormal volume profiles

Full-sample event-fixed-effects estimates by window (standardized abnormal log volume): hardware 0.033 ($[-10, -2]$, n.s.), 0.121*** $[0, 1]$, 0.086*** $[0, 5]$, 0.074*** $[0, 10]$, 0.046** $[0, 20]$; deployer never exceeds $|0.025|$ (max $|t|$ at $[0, 20]$, $p = 0.10$); competitor turns negative at long windows (-0.067 ***, $[0, 20]$); cloud 0.292*** at $[0, 1]$ (0.435*** in 2025–26); owner 0.386** at $[0, 1]$. Pre-event windows show no abnormal volume for hardware (0.033, $p = 0.22$), deployers, integrators, or competitors; the exception is cloud, whose abnormal volume is already 0.203 ($p = 2.5 \times 10^{-6}$, FDR-robust) over $[-10, -2]$.

The hardware pre-event price run-up is thus not accompanied by trading intensity, while cloud names churn both before and at announcements without directional pricing.

Appendix E.4. Composition of the hardware premium

Dropping any single hardware firm leaves the full-sample coefficient between 0.0118 and 0.0163 (23 leave-one-out fits); excluding NVIDIA specifically yields 0.0153 ($p = 0.023$). The premium is also not a pure semiconductor-index exposure: splitting hardware into SOX members and non-SOX hardware names (server, networking, and storage suppliers) gives 0.0122 and 0.0275 respectively, with the difference significant at $p = 0.036$ —though the non-SOX group counts only five firms, so this contrast is suggestive.

Appendix E.5. Position levels by phase (event fixed effects)

Hardware: 2024 -0.0253 (0.0112)**, 2025 0.0394 (0.0073)***, 2026 0.0309 (0.0263). Integrator: 2024 -0.0013 (0.0077), 2025 -0.0071 (0.0055), 2026 -0.0045 (0.0402). Deployer: 2024 -0.0018 (0.0044), 2025 0.0029 (0.0034), 2026 -0.0172 (0.0097). The 2026 cells rest on nine events. The 2024 hardware negative decomposes by modality as in Table 4: LLM events -0.0150 (0.0137), media events -0.0468 (0.0194)**; its media component flips sign in 2025–26 ($+0.0346$ **), while the LLM component’s formation is monotone.

Appendix E.6. Capability grid

Beyond Table 5: capability-surprise (index minus running pre-event frontier maximum) behaves like the level (pooled -0.00007 , $p = 0.73$; within closed -0.00076 , $p = 0.025$); a frontier-breaking dummy is n.s. (-0.0059 ,

$p = 0.17$); related-firms-only slope -0.00015 ($p = 0.47$); related $\times$ closed -0.00047 ($p = 0.20$). Period split of the closed slope: 2022–24 -0.00014 ($p = 0.87$), 2025–26 -0.00058 ($p = 0.043$). The eight highest-index closed events (GPT-5.x, Claude 4.5/4.6 tier, Gemini 3) all carry *positive* mean CARs ($+0.1\%$ to $+2.9\%$)—the negative slope is a within-year anticipation gradient, not losses on flagships.

Appendix E.7. Calendar-time portfolio

Long 30 hardware / short 40 downstream, equal-weighted, April 2022–April 2026 (1,023 days): three-factor daily alpha 9.7bp ($p = 0.054$; annualized $\approx 24\%$). Event-window attribution fails: windows cover 56% of days; the in-window increment is +10.8bp ($p = 0.26$); the dose-response in concurrent event count is humped (0 events: 14%; 1: 48%[†]; 2: 54%[†]; 3+: 17% annualized); the 2025–26 subperiod is high in and out of windows. Use as magnitude only.

Appendix E.8. Reconciliation with an independent specification search

An independently run 3,144-specification search (same panel, HC0 event clustering, event fixed effects, FDR control) reproduces the hardware–time findings (hardware $\times$ trend positive in 100% of specifications, family $q = 0.002$) and proposes downstream repricing claims (deployer positive in 2024, negative post-2025) that do *not* survive our preferred design. The discrepancy is diagnostic, not arbitrary: the search pools all phases and lets only the focal dimension vary by phase, holding the other six position coefficients constant across periods; since the hardware effect itself swings from -2.5pp (2024) to $+3.9\text{pp}$ (2025), the constrained design leaks misspecification into

focal-dimension phase terms, and HC0 understates CR2 uncertainty at 124 clusters. Under phase-by-phase subsample estimation with all coefficients free and CR2, the downstream cells are zeros (2024 deployer -0.0018 ; 2025 integrator -0.0071 , $p = 0.20$). Both designs are reported for transparency; the paper rests on the unconstrained one.

Appendix E.9. The 2024 hardware negative: forensics

The cell survives benchmark changes (FF3 -0.0280^{**} , QQQ -0.0266^* , SOXX -0.0190^{**}), NVDA exclusion (-0.0273^{**}), jackknife over events (range $[-0.029, -0.022]$), and pre-CAR control (-0.0235^*); the pre-event window is also negative (-0.0199^{**}), indicating release-dense periods coincided with hardware-weak stretches in 2024. It does not survive restriction to compute-relevant events: LLM-only -0.0150 ($p = 0.28$), and with pre-CAR control -0.0136 ($p = 0.33$). Interpretation in the text: formation is monotone for LLM events; the media-event component reflects 2024’s rotation episodes and flips positive once video generation became compute-intensive.